JetStock

Inventory & Product Management Application

Capstone Project | Software Engineering Program | April 2026

Products

Full CRUD

Inventory

Live tracking

Suppliers

Contacts

The Problem & Our Solution

❌ The Problem

- Manual tracking via notebooks & spreadsheets
- No real-time stock level visibility
- Missing low-stock alerts until out of stock
- No central system — disorganised records
- Existing tools are too complex & expensive
- Time wasted searching for product info

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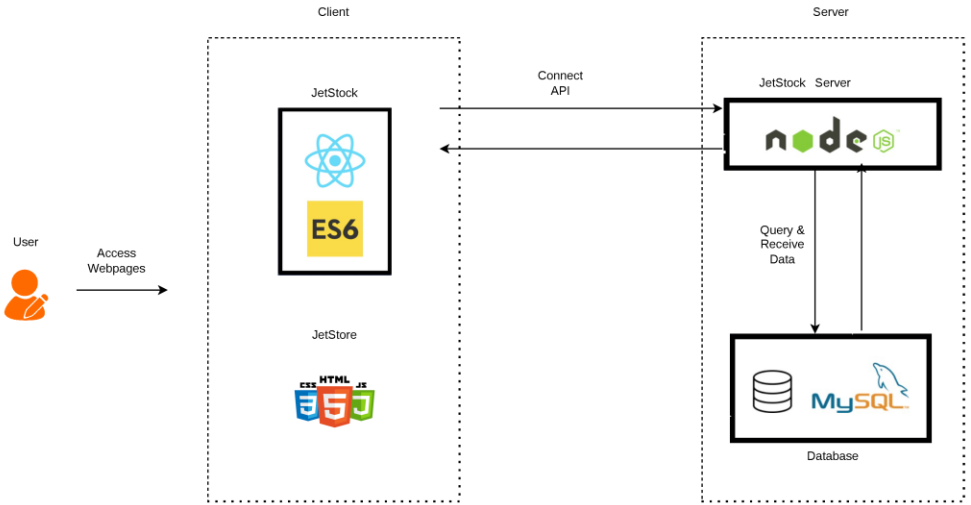
✅ JetStock Solution

- Central web app for all inventory data
- Live stock quantity tracking
- Low-stock alerts & reorder level visibility
- Organised by product, category & supplier
- Simple, lightweight — built for small business
- Fast search & filter for any product

System Architecture & Tech Stack



Key Features

Authentication

JWT-based login & registration with role-based access

Product Management

Full CRUD for products with SKU, price, category & supplier

Categories

Organise products into categories for easy filtering

Suppliers

Manage supplier contacts linked to your products

Inventory Control

Real-time quantity tracking with low-stock & out-of-stock alerts

Stock Movements

Log stock IN/OUT with movement type: manual, online, return, damage

Orders

Create and track customer orders with full order item management

Dashboard

Summary view of total products, low-stock count and inventory stats

Database Design (ERD)

Users

id, first_name, last_name, email, role

Products

id, sku, name, price, supplier_id, category_id

Categories

id, name, description, user_id

Stock_movements

id, product_id, movement_type, source_type, qty

Inventory

id, product_id, quantity_on_hand, reorder_level

Suppliers

id, name, contact_person, email, phone

Orders

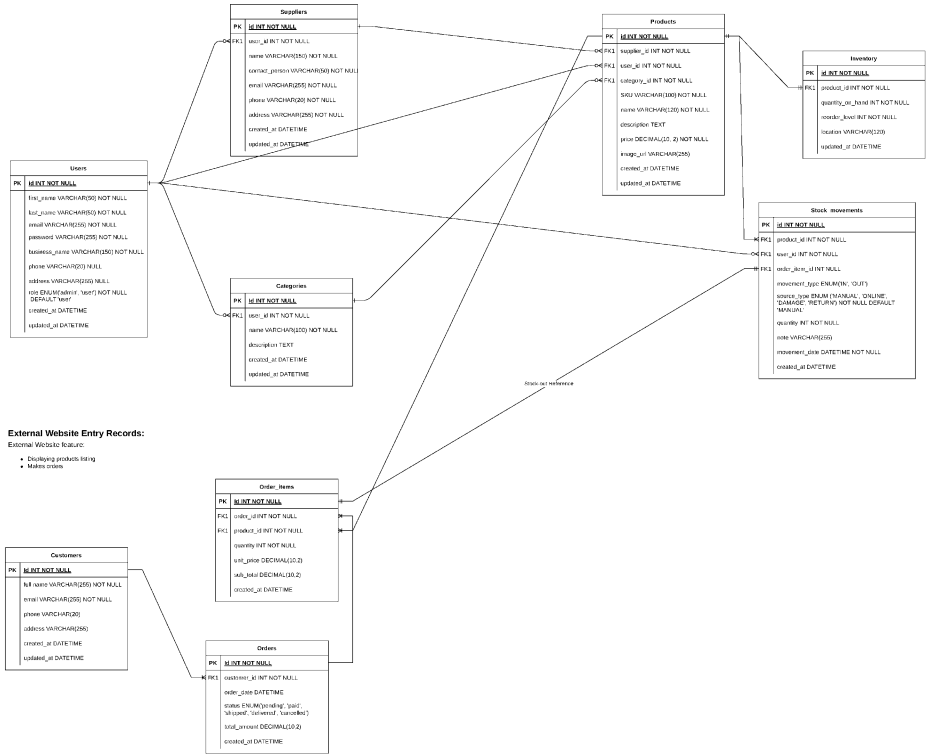
id, customer_id, status, total_amount

Order_items

id, order_id, product_id, quantity, unit_price

Customers

id, full_name, email, phone, address



External Website Entry Records:

External Website feature:

- Displaying products listing
- Makes orders

Project Planning & Timeline

Phase 1 Mar 13–15	Phase 2 Mar 16–24	Phase 3 Mar 24–28	Phase 4 Mar 28–31
Planning & Design	**Backend Dev**	**Frontend Dev**	**Polish & Delivery**
✓ Define project ✓ User stories ✓ Architecture diagram ✓ Lo-fi wireframes	✓ Models & DB setup ✓ CRUD APIs ✓ Auth / JWT ✓ Error handling ✓ Unit tests	✓ Login/Signup ✓ Dashboard ✓ All listing pages ✓ Forms & search ✓ Connect API	✓ UI polish ✓ Frontend tests ✓ Documentation ✓ PPT slides

Testing Strategy

Unit Testing (Jest)

- CRUD controller tests
- Stock movement logic
- Auth middleware validation
- Error handling responses

API Testing (Postman)

- All REST endpoints verified
- Valid & invalid request bodies
- Auth headers & protected routes
- HTTP status code verification

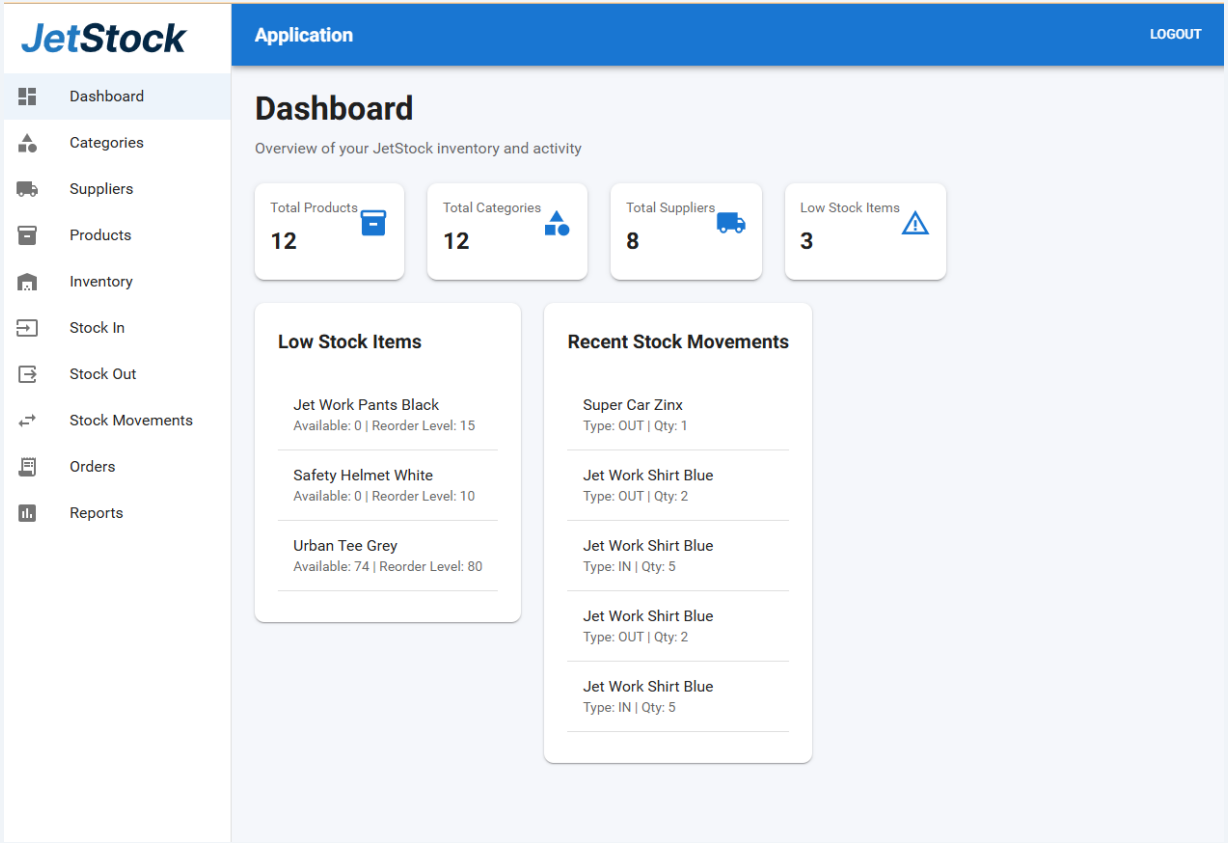
Manual UI Testing

- Login/Registration flow
- All CRUD form validation
- Dashboard real-time data
- Search, filter & pagination

Edge Cases Handled

- Stock out > available qty
- Duplicate SKU detection
- Delete with linked records
- Out-of-stock visual alerts

Application Walkthrough



Project Outcomes

What JetStock delivered

- ✅ Full-stack web app with React + Node.js + MySQL
- ✅ Complete CRUD for Products, Categories, Suppliers & Orders
- ✅ Real-time inventory tracking with low-stock alerts
- ✅ JWT authentication with protected routes
- ✅ External storefront (JetStore) connected via shared API
- ✅ Unit tests (Jest) + API tests (Postman)

https://github.com/mensonsundash/jetstock-capstone